



11751 - Increasing the sensitivity of all high-contrast imaging through physics-based PSF subtraction and automatic differentiation

Cycle: 5, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

We propose to re-process all exoplanet direct-imaging observations (NIRCam and MIRI) with a new physics-based PSF subtraction algorithm enabled by machine learning optimization tools to increase the exoplanet detection sensitivity. Early results of this method on NIRCam data have shown improvements by at least a factor of 3 in exoplanet detection sensitivity relative to traditional algorithms. This proposal will 1) re-process all high-contrast NIRCam observations, particularly those aiming to image young, low mass giant planets for the first time (4050, 5835, 6005) and increase their sensitivity to mass ranges and orbital separations previously inaccessible, 2) extend the method to MIRI to re-process high-contrast MIRI observations with improved sensitivity and higher precision astrometry and photometry, and 3) build an open-source tool to make this framework available to the community, paving the way its application in future JWST high-contrast imaging and other JWST modes.

OBSERVING DESCRIPTION